

Name: Avinash M
Register No: 192521196
Course: Computer Network
Course Code: CSA0707

Concept Maps: Error Control, OSI Layers, and Network Troubleshooting

Question 4: Error Control Across Layers

Concept Map:

	ERROR CONTROL	
	↓	
Data Link Layer		Transport Layer
↓		↓
Error Detection (checks for errors)		Error Correction (fixes errors)

Analysis: How the Layers Work Together for Reliability

Error control is not handled by a single layer of the network. Instead, two different layers each play a distinct, complementary role, and together they make data transmission far more reliable than either could alone.

1. Data Link Layer – Error Detection

The Data Link Layer is responsible for moving data across a single physical link (for example, one cable segment or one Wi-Fi hop). Since electrical noise, interference, or signal weakening can corrupt bits during this short hop, this layer attaches a checksum or CRC (Cyclic Redundancy Check) value to every frame.

- Purpose: to detect whether a frame arrived damaged.
- Scope: local, hop-by-hop — it only checks the link the frame just crossed.
- Action on error: if damage is detected, the frame is usually just discarded (this layer typically detects, not fixes).

2. Transport Layer – Error Correction

The Transport Layer (for example, TCP) is responsible for the entire end-to-end journey of data, from the original sender to the final receiver, possibly across many links and routers. It uses sequence numbers, acknowledgments (ACKs), and retransmission timers to recover from problems.

- Purpose: to correct/recover from errors or loss, not just detect them.
- Scope: end-to-end — it does not care how many links the data crossed, only whether it arrived correctly overall.

- Action on error: missing or corrupted data is retransmitted until it is confirmed received correctly.

3. Why the Combination Improves Reliability

- Layered defense: a damaged frame is caught early (Data Link) so it never wastes time travelling further; anything that still slips through or gets lost across the network is caught late (Transport).
- Efficiency: local retransmission at the Data Link layer (on wireless links, for instance) fixes small local problems quickly, without forcing the entire end-to-end connection to restart.
- Completeness: the Data Link layer alone cannot guarantee the data survives multiple hops and routers; the Transport layer alone would be slow if it had to catch every small local glitch. Together they cover both the small-scale and large-scale sources of error.
- Result: the application receives data that has been checked at the local level and guaranteed at the global level — this two-tier checking is what makes protocols like TCP/IP reliable even over imperfect physical networks.

Summary: Data Link Layer detects local, per-link errors; Transport Layer corrects errors across the entire end-to-end path. Their combination gives both fast local error-catching and guaranteed overall delivery.

Question 5: Concept Map for Web Browsing Using OSI Layers

Concept Map (data flows downward as the browser sends a request, and upward in reverse when the reply is received):

7. Application Layer	Browser (HTTP/HTTPS) requests the webpage
↓	
6. Presentation Layer	Formats/encrypts data (e.g. TLS/SSL, encoding)
↓	
5. Session Layer	Manages the session between browser and server
↓	
4. Transport Layer	TCP breaks data into segments, ensures reliable delivery
↓	
3. Network Layer	IP addressing and routing packets to the server
↓	
2. Data Link Layer	Frames packets for the local network (e.g. Ethernet/Wi-Fi)
↓	
1. Physical Layer	Sends raw bits as electrical/radio/light signals

Analysis: Effect of a Failure at Each Layer

Because the OSI layers form a dependency chain, each layer relies completely on the layer below it to move data, and provides a service to the layer above it. A failure at any single layer breaks the chain and stops web browsing, but the way it fails looks different depending on where the break occurs.

Failure Examples, Layer by Layer

- Physical Layer fails (e.g. unplugged cable, Wi-Fi radio off): No signal at all reaches the network. The browser cannot connect; the OS usually reports "no network connection."
- Data Link Layer fails (e.g. wrong Wi-Fi password, MAC filtering, NIC driver crash): The device cannot exchange frames with the local switch/router, so it cannot even reach its own local network, let alone the internet.
- Network Layer fails (e.g. incorrect IP configuration, routing failure, DNS issue often surfaces here too): Packets cannot be routed to the destination server. The browser typically shows a timeout or "server not found" error.
- Transport Layer fails (e.g. firewall blocking the port, TCP handshake failure): A route to the server may exist, but no reliable connection/session can be established, so the page never starts loading.
- Session/Presentation Layer fails (e.g. TLS/SSL certificate error, encoding mismatch): The connection reaches the server, but the browser refuses or cannot interpret the response — this is the familiar "your connection is not private" warning.
- Application Layer fails (e.g. the web server itself is down, or returns an HTTP error): The network path works perfectly, but the requested service is not available — this is where a "404 Not Found" or "500 Internal Server Error" comes from.

Why One Failure Breaks the Whole Chain

Every layer packages the data it receives from the layer above and hands it to the layer below (this packaging is called encapsulation). If any single layer cannot perform its job, the layers above it have no way to reach the destination, because they depend entirely on the lower layer to physically move their data. This is why troubleshooting network problems typically starts at the Physical Layer and moves upward — confirming each layer works before assuming a higher layer is at fault.

Summary: Web browsing depends on all seven OSI layers working in sequence. A failure at a lower layer (Physical, Data Link, Network) prevents any connection from forming, while a failure at a higher layer (Session, Presentation, Application) allows a connection but breaks the actual browsing experience.

Question 6: Troubleshooting Using a Layered Concept Map

Concept Map (✓ = working, ✗ = not working):



Analysis: Locating the Fault

This concept map is showing the status of each layer along the path of a single connection attempt. Reading it from left (Physical) to right (Application) reveals exactly where the chain of communication breaks:

- Physical Layer (✓): The cable, Wi-Fi signal, or network interface card is working — the device has a physical connection to the network.

- Data Link Layer (✓): Frames are being exchanged successfully with the local switch or access point — the device is properly connected to its local network segment.
- Network Layer (✗): This is the first layer that fails. Packets cannot be correctly routed — likely causes include a missing/incorrect IP address, a misconfigured default gateway, wrong subnet mask, or a routing failure.
- Transport Layer (✗): Naturally fails as a consequence — no TCP/UDP session can be established if packets are not even being routed.
- Application Layer (✗): Also fails as a consequence — the web browser, email client, or any application has nothing to work with, since no data is reaching the destination at all.

Why the Concept Map Helps with Troubleshooting

- Pinpoints the root cause: instead of guessing, the map shows the fault begins exactly at the Network Layer — the two ✗ marks after it are effects, not separate causes.
- Saves time: without this layered view, someone might waste time checking the web browser (Application Layer) or restarting apps, when the real problem is IP/routing configuration.
- Follows a logical, bottom-up method: this mirrors the real-world network troubleshooting approach — verify the physical connection first, then the local link, then addressing/routing, then the transport session, and finally the application — stopping at the first layer that fails.
- Predicts downstream impact: it also explains why higher layers fail automatically once a lower layer breaks, so the ✗ marks at Transport and Application do not need separate investigation once Network Layer is fixed.

Summary: Because Physical and Data Link both work, the local connection is fine. The failure starts at the Network Layer (likely an IP/routing/addressing problem), and this single fault is the reason both Transport and Application also fail — fixing the Network Layer issue should restore the entire connection.